Физические величины

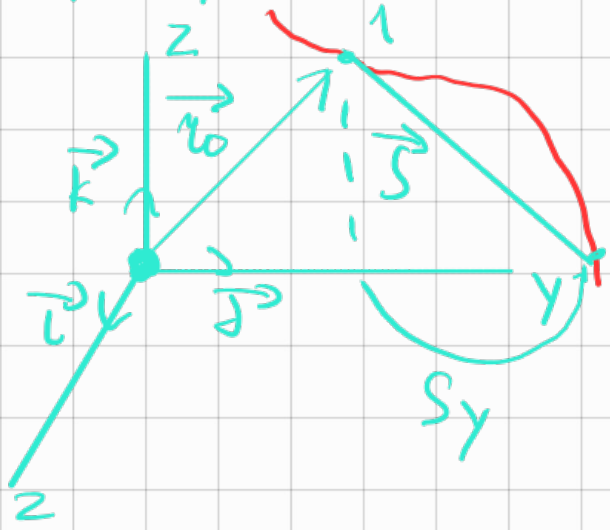
- Скалярные (const)
- Векторные
- Тензорные

Нельзя приравнивать число к векторам.

$$6\ \frac{км}{ч^2} = 6 \cdot \frac{1000\ м}{3600\ с \cdot 3600\ с} = 4{,}6 \cdot 10^{-4}\ \frac{м}{с^2}$$

Абсолютная погрешность – в числах
Относительная – в %

Движение – изменение положения тела в пространстве относительно других тел.

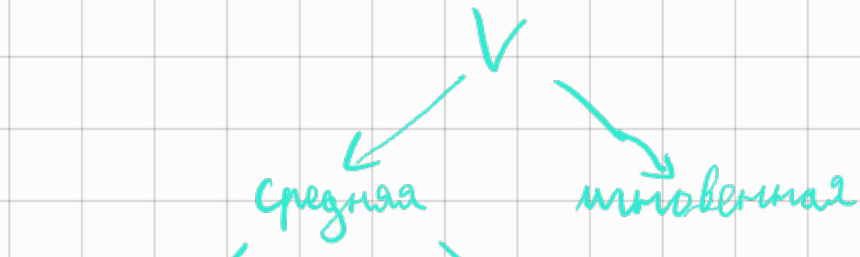


1) x_0 y_0 z_0

2) $\vec{r_0} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}$

$\vec{r_0} = x_0\vec{i} + y_0\vec{j} + z_0\vec{k}$

$$\vec{r} = \vec{r_0} + \vec{S}$$



$\vec{V}_{ср} = \frac{\vec{S}}{\Delta t} = \frac{\Delta \vec{r}}{\Delta t}$ $\quad V_{пут} = \frac{l}{\Delta t}$ $\quad \vec{S} = \vec{r} - \vec{r}_0 = \Delta \vec{r}$



$\vec{a}_{ср} = \frac{\Delta \vec{V}}{\Delta t}$

ПРАКТИКА

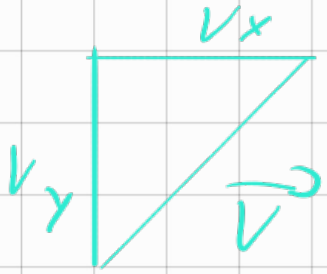
$\vec{r} = (3 - t^2)\vec{i} + 4t\vec{j}$

$t = 2c$
$V - ?$
$a - ?$
$S_{3-} - ?$

$$\begin{cases} x = 3 - t^2 \\ y = 4t \end{cases}$$

$\vec{V} = \vec{r}'(t) = -2\vec{i}$

$$\begin{cases} V_x = \frac{\partial x}{\partial t} = -2t \\ V_y = \frac{\partial y}{\partial t} = 4 \end{cases}$$

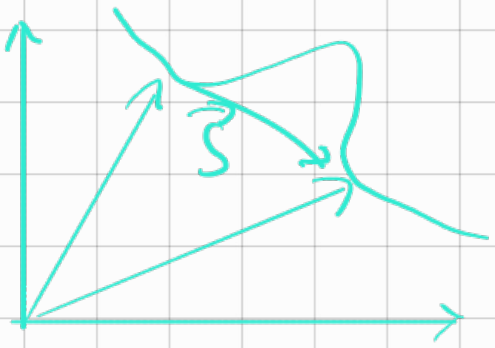


$V = \sqrt{V_x^2 + V_y^2}$

$$a = \frac{\Delta v}{\Delta t} = -2\vec{i}$$

$$\begin{cases} a_x = -2 \\ a_y = 0 \end{cases}$$

$$a = \sqrt{a_x^2 + a_y^2} = 2$$



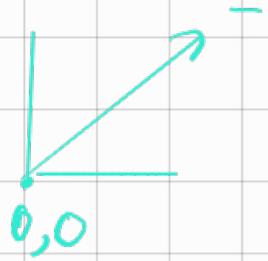
$$\Delta\vec{r} =$$

$$\vec{r_3} = (3-9)\vec{i} + 12\vec{j}$$

$$\vec{r_2} = -\vec{i} + 8\vec{j}$$

$$\Delta\vec{r_{23}} = -5\vec{i} + 4\vec{j}$$

$$S = \Delta r = \sqrt{25 + 16} = \sqrt{41}$$



$$x_1 = \overbrace{3t^2 - 7t + 6}^{S} \quad x_0$$

$$x_2 = 3t^2 - 7t + 1 \quad x_0$$

$$\Delta r = S = x_{кон} - x_0$$

$$a = 6$$

$$x = x_0 + S$$

$$\Delta V = \Delta S' = 6t - 7 \quad v_0$$

$$a = \frac{dV}{dt} = 6$$

$$a = \frac{dv}{dt} \Rightarrow d\vec{V} = \vec{a}\,dt$$

$$\vec{V} = \int \vec{a}\,dt$$

$$a = 6x^2 - c^4$$

$$\vec{V} = \frac{d\vec{S}}{dt} \Rightarrow d\vec{S} = \vec{V} \cdot dt$$

$$V = (12t - 3t^2)\ \frac{м}{с}$$

остановка $V = 0$; $12t - 3t^2 = 0$

$t(12 - 3t) = 0$

$t_1 = 0$ $\quad t_2 = 4c$

$$S = \int_0^4 V\,dt =$$

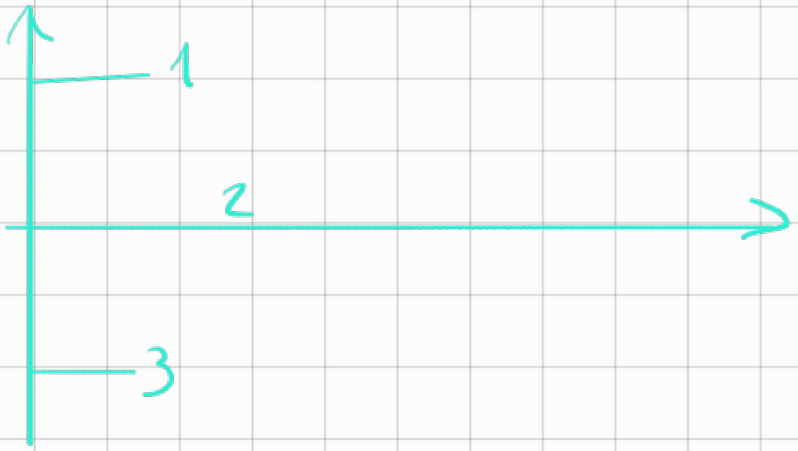
$$\int_0^4 (12t - 3t^2)\,dt = \left(\frac{12 \cdot t^2}{2} - \frac{3t^2}{3}\right)\Big|_0^4$$

$$6t^2 - t^2\Big|_0^4 = 32\,м$$

1. $P \parallel \Pi_1$

$\vec{V} = const$

$V = \frac{S}{t} = \frac{x - x_0}{t}$



2. РУД

$a = \text{Const}$ $\qquad a = \frac{v - v_0}{t} = \frac{\Delta v}{\Delta t}$

$v = v_0 + at$

$v = \int a \, dt = \int a \cdot t^0 \, dt = at + v_0$

$x = x_0 + s = x_0 + v_0 t + \frac{at^2}{2}$